THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 8 · SET 04

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — **1 questions · 12 marks**

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The square root of 0.0081 is:

(a) 0.09 (b) 0.9 (c) 0.009 (d) 0.081

Ans: (a) 0.0081 = 81/10000 ⇒ √ = 9/100 = 0.09.

(ii) The cube of −4 is:

(a) 64 (b) −64 (c) −16 (d) 16

Ans: (b) $(-4)^3 = -64$.

(iii) The number of non-square numbers between 6^2 and 7^2 is:

(a) 6 (b) 13 (c) 24 (d) 12

Ans: (d) Between n^2 and $(n+1)^2$ there are 2n non-squares: 2 × 6 = 12.

(iv) The value of $2^{-3} \times 2^5$ is:

(a) 2 (b) 8 (c) 1/4 (d) 4

Ans: (d) 2^(−3+5) = $2^2 = 4$.

(v) The Roman numeral XLIV stands for:

(a) 46 (b) 64 (c) 44 (d) 36

Ans: (c) XL = 40 and IV = 4 ⇒ 44.

(vi) $(101)_2$ in the decimal system is:

(a) 3 (b) 5 (c) 6 (d) 7

Ans: (b) 1×4 + 0×2 + 1 = 5.

(vii) A quadrilateral with all sides equal and one angle 90° is a:

(a) rhombus that is not a square (b) square
(c) kite (d) trapezium

Ans: (b) A rhombus with a right angle is a square.

(viii) The diagonals of every parallelogram:

(a) are equal (b) are perpendicular
(c) bisect each other (d) are parallel

Ans: (c) Basic property of a parallelogram.

(ix) $(a + b)^2 - (a - b)^2$ equals:

(a) 2ab (b) $2a^2 + 2b^2$ (c) 0 (d) 4ab

Ans: (d) $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

(x) 5 kg : 500 g in simplest form is:

(a) 1 : 10 (b) 5 : 500 (c) 10 : 1 (d) 1 : 100

Ans: (c) 5000 g : 500 g = 10 : 1.

(xi) $x^2 - 49$ can be factorised as:

(a) $(x - 7)^2$ (b) $(x + 7)^2$

(c) $(x - 7)(x + 7)$ (d) $(x - 49)(x + 1)$

Ans: (c) $a^2 - b^2 = (a - b)(a + b)$.

(xii) If 3 : 5 :: x : 20, then x equals:

(a) 15 (b) 8 (c) 10 (d) 12

Ans: (d) $5x = 60 \Rightarrow x = 12$.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 2025 by prime factorisation.

$2025 = 3^4 \times 5^2$ (1 mark)
$\sqrt{2025} = 3^2 \times 5 = 45$ (1 mark)

3. Is 392 a perfect cube? If not, find the smallest number by which it must be multiplied to get a perfect cube.

$392 = 2^3 \times 7^2$; 7 appears twice, not a multiple of 3 times, so not a perfect cube (1 mark)
Multiply by 7: $2744 = 14^3$ (1 mark)

4. Write 53,000,000 and 0.000345 in standard form.

5.3×10^7 (1 mark)
3.45×10^{-4} (1 mark)

5. Write 2024 in Roman numerals.

2000 = MM, 20 = XX, 4 = IV (1 mark) ⇒ MMXXIV (1 mark)

OR

Convert $(234)_5$ into a decimal number.

2×25 + 3×5 + 4 (1 mark) = 69 (1 mark)

6. In rhombus ABCD, ∠A = 70°. Find ∠B and ∠C.

Adjacent angles are supplementary: ∠B = 110° (1 mark)
Opposite angles are equal: ∠C = ∠A = 70° (1 mark)

7. Expand $(x + 4)^2$ and $(3x + 2)(3x - 2)$.

$(x + 4)^2 = x^2 + 8x + 16$ (1 mark)
$(3x + 2)(3x - 2) = 9x^2 - 4$ (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the square root of 1.69 and the cube root of 0.512.

$\sqrt{1.69} = \sqrt{(169/100)} = 13/10 = 1.3$ (1½ marks)
$\sqrt[3]{0.512} = \sqrt[3]{(512/1000)} = 8/10 = 0.8$ (1½ marks)

9. The angles of a quadrilateral are in the ratio 2 : 3 : 3 : 4. Find each angle.

Total parts = 12; 1 part = 360°/12 = 30° (1 mark)
Angles: 60°, 90°, 90°, 120° (2 marks)

OR

In parallelogram PQRS, ∠P : ∠Q = 2 : 3. Find all four angles.

> ∠P + ∠Q = 180° ⇒ 5 parts = 180°, 1 part = 36° (1 mark)
> ∠P = 72° = ∠R, ∠Q = 108° = ∠S (2 marks)

10. Using identities, evaluate: (i) $(1.02)^2$ (ii) 48×52 (iii) 302^2

> (i) $(1 + 0.02)^2 = 1 + 0.04 + 0.0004 = 1.0404$ (1 mark)
> (ii) $(50 - 2)(50 + 2) = 2500 - 4 = 2496$ (1 mark)
> (iii) $(300 + 2)^2 = 90000 + 1200 + 4 = 91204$ (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 288 must be multiplied so that the product is a perfect square. Find the square root of the product.
(ii) Find the square root of 5476 by prime factorisation.

> (i) $288 = 2^5 \times 3^2$ (1 mark); multiply by 2 (1 mark); $576 = 24^2 \Rightarrow \sqrt{} = 24$ (1 mark)
> (ii) $5476 = 2^2 \times 37^2$ (1 mark) ⇒ $\sqrt{} = 2 \times 37 = 74$ (1 mark)

OR

(i) Find the smallest number by which 1944 must be divided so that the quotient is a perfect cube. Find the cube root of the quotient.
(ii) The volume of a cube is 0.064 m^3. Find the length of its edge.

> (i) $1944 = 2^3 \times 3^5$ (1 mark); divide by $3^2 = 9$ (1 mark); quotient $216 = 6^3 \Rightarrow$ cube root 6 (1 mark)
> (ii) $0.064 = 64/1000 = (4/10)^3 \Rightarrow$ edge = 0.4 m (2 marks)

12. Simplify: (i) $5^{-3} \times 5^5$ (ii) $(2/3)^{-3}$ (iii) $7^6 \div 7^8$ (iv) $(x^{-2}y^3)^2$ (v) $3^o + 4^{-1}$

> (i) $5^2 = 25$ (ii) 27/8 (iii) $7^{-2} = 1/49$ (iv) y^6/x^4 (v) 1 + 1/4 = 5/4 (1 mark each)

OR

(i) Write 1994 in Roman numerals. (ii) Convert 45 to base 2. (iii) Convert $(212)_3$ to decimal. (iv) Add $(110)_2$ and $(111)_2$. (v) Convert 58 to base 8.

> (i) MCMXCIV (ii) $45 = 32 + 8 + 4 + 1 = (101101)_2$ (iii) $2\times9 + 1\times3 + 2 = 23$ (iv) $6 + 7 = 13 = (1101)_2$ (v) $58 = 7\times8 + 2 = (72)_8$ (1 mark each)

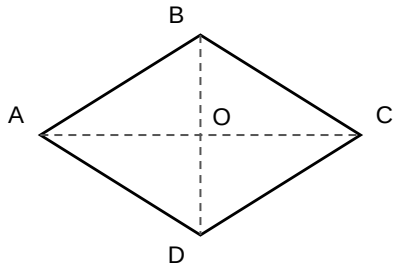
13. (i) Divide ₹ 2,400 among A, B and C in the ratio 3 : 5 : 4.
(ii) 12 workers finish a job in 15 days. How many days will 20 workers take?
(iii) Find x if 4 : 9 :: 16 : x.
(iv) A train travels 240 km in 4 hours. How long will it take for 420 km at the same speed?
(v) A 40-litre mixture has milk and water in the ratio 7 : 3. Find the quantity of water.

> (i) 12 parts, 1 part = ₹ 200 ⇒ ₹ 600, ₹ 1000, ₹ 800 (1 mark)
> (ii) Inverse variation: 12 × 15 = 20 × d ⇒ d = 9 days (1 mark)
> (iii) 4x = 144 ⇒ x = 36 (1 mark)
> (iv) 60 km/h ⇒ 420 ÷ 60 = 7 hours (1 mark)
> (v) Water = 3/10 × 40 = 12 litres (1 mark)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A rhombus-shaped garden ABCD has diagonals AC = 16 m and BD = 12 m meeting at O. Each side of the garden is 10 m. ∠ABC = 100°.



(i) What is the angle between the diagonals AC and BD?

> The diagonals of a rhombus are perpendicular: 90°

(ii) Find the length of AO.

> Diagonals bisect each other: AO = 16/2 = 8 m

(iii) Find the cost of fencing the garden at ₹ 50 per metre.

> Perimeter = 4 × 10 = 40 m (1 mark); cost = 40 × 50 = ₹ 2000 (1 mark)

OR

(iii) Find ∠BAD and ∠OAB.

> ∠BAD = 180° − 100° = 80° (1 mark); diagonal AC bisects ∠A ⇒ ∠OAB = 40° (1 mark)

15. In a laboratory, a bacterial culture doubles every hour. Starting with 1 bacterium, the number after n hours is 2^n.

(i) How many bacteria are there after 5 hours?

> $2^5 = 32$

(ii) How many bacteria are there after 10 hours?

> $2^{10} = 1024$

(iii) After how many hours will the count reach 4096?

> $4096 = 2^{12}$ (1 mark) ⇒ 12 hours (1 mark)

OR

(iii) How many times more bacteria are there at 12 hours than at 9 hours?

> $2^{12} \div 2^9 = 2^3$ (1 mark) = 8 times (1 mark)

16. A fruit punch is made by mixing orange juice, pineapple juice and soda in the ratio 4 : 3 : 1. A party needs 4 litres of punch.

(i) What is the total number of parts in the mixture?

> 4 + 3 + 1 = 8

(ii) How much orange juice is needed for 4 litres of punch?

> 1 part = 4/8 = 0.5 L; orange juice = 4 × 0.5 = 2 L

(iii) How much soda is needed to make 10 litres of the same punch?

> 1 part = 10/8 = 1.25 L (1 mark); soda = 1 part = 1.25 L (1 mark)

OR

(iii) Only 4.5 litres of pineapple juice is available. What is the maximum amount of punch that can be made keeping the same ratio?

> 3 parts = 4.5 L ⇒ 1 part = 1.5 L (1 mark); total = 8 × 1.5 = 12 L (1 mark)